

Week 1 - Python Basics - Personal Introduction Program

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1. Project Overview

This project was developed as part of Week 1 of my Data Science Internship at The Developers Arena.

The Personal Introduction Program is a beginner-friendly Python program that collects the user's name, age, and favorite hobby using keyboard input and displays a personalized welcome message.

The project demonstrates fundamental Python programming concepts such as variables, input(), print(), strings, f-strings, and basic program execution.

2. Objectives

The main objectives of this project are:

- To understand the basics of Python programming.
- To learn how to use variables to store information.
- To collect user information using input().
- To display information using print().
- To understand basic string formatting using f-strings.
- To create and run a simple Python program.
- To practice writing clean and readable beginner-level Python code.

3. Technologies Used

- Python 3.x
- Visual Studio Code
- Git
- GitHub

No external Python libraries are required for this project.

4. Setup Instructions

Prerequisites

Python 3.x should be installed on the system.

To verify the Python installation, open the VS Code terminal or PowerShell and run:

```
python --version
```

Project Setup

1. Clone or download the GitHub repository.
2. Open the project folder in Visual Studio Code.
3. Navigate to the Task_1_Python_Basics folder.

4. Make sure the following files are present:

Task_1_Python_Basics/

- personal_intro.py
- README.md
- requirements.txt

5. How to Run the Program

Open the terminal inside the Task_1_Python_Basics folder and run:

```
python personal_intro.py
```

The program will ask the user for:

- Name
- Age
- Favorite hobby

After entering the information, the program displays a personalized welcome message.

6. Code Structure

The main Python file is:

personal_intro.py

The program follows a simple step-by-step structure:

1. Ask the user for their name.
2. Store the name in a variable.
3. Ask the user for their age.
4. Store the age in a variable.
5. Ask the user for their favorite hobby.
6. Store the hobby in a variable.
7. Display a personalized welcome message using `print()` and an f-string.

The project also contains:

README.md

- Contains project information, setup instructions, usage details, and learning outcomes.

requirements.txt

- Contains dependency information. No external dependencies are required.

7. Technical Details

The project meets the technical requirements specified for the task.

input()

The input() function is used to collect information from the user.

Variables

Variables are used to store the user's name, age, and hobby.

print()

The print() function is used to display the welcome message.

String Formatting

An f-string is used to insert the user's information into the personalized welcome message.

Beginner-Friendly Code

The program uses simple Python syntax and does not require any external libraries or advanced programming concepts.

8. Example Input and Output

Example input:

Name: Siddhesh

Age: 21

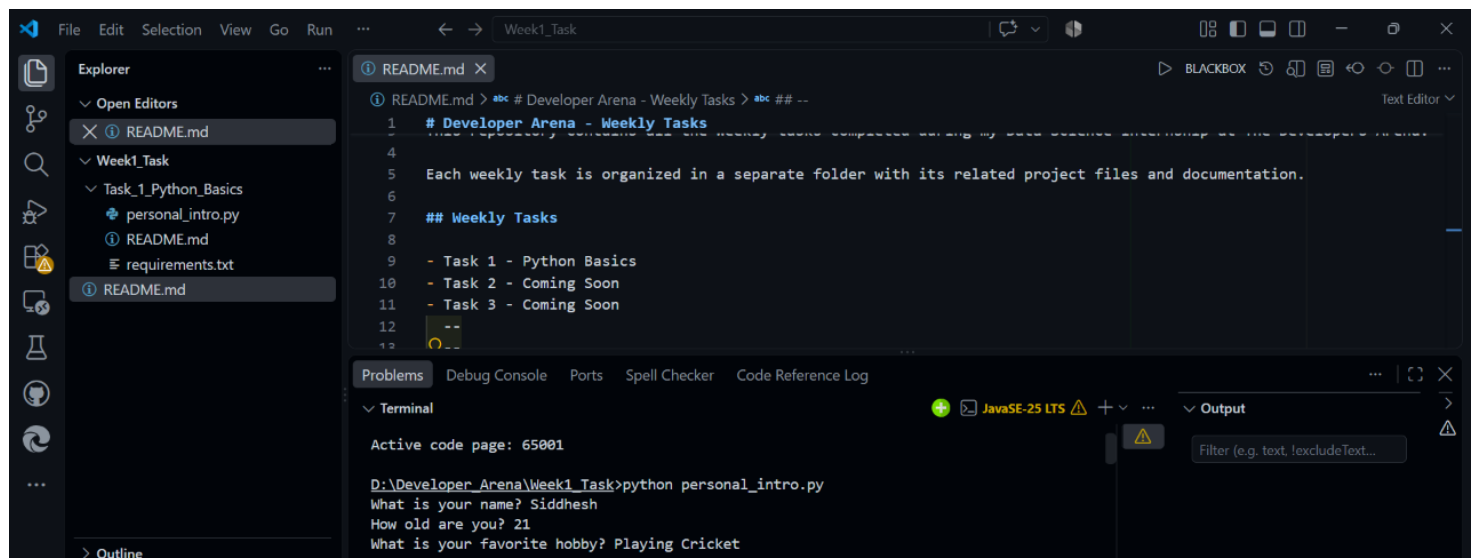
Hobby: Coding

Example output:

💎 Welcome Siddhesh! 💎

You are 21 years old and love Coding.

9. Visual Documentation



10. Testing Evidence

The program was tested by running:

python personal_intro.py

Test case:

Name: Siddhesh

Age: 21

Hobby: Coding

Expected result:

The program should display a personalized welcome message containing the entered name, age, and hobby.

Actual result:

The program successfully displayed the personalized welcome message.

[Insert screenshot of successful test output here.]

11. What I Learned

Through this project, I learned the fundamentals of Python programming.

I learned how to:

- Create and run a Python program.
- Use variables to store values.
- Take user input using `input()`.
- Display information using `print()`.
- Work with strings.
- Use f-strings for personalized output.
- Organize a small Python project.
- Test a Python program and verify its output.
- Maintain project files using Git and GitHub.

12. Conclusion

The Personal Introduction Program provided practical experience with basic Python programming concepts. It helped me understand how user input, variables, strings, and output work together to create a simple interactive program.